

## Entry: 1

### Item: 4

#### Question:

Now run with these

flags: -l 4:100,1:0. These

flags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be done. Can you find out if you were right.

#### Answer:

It takes 10 units of time to complete both processes.

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE state.)

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8.0

#### Score 2:

8.0

#### Score 3:

8.0

#### Score outlier:

16

## Entry: 2

### Item: 4

#### Question:

Now run with these flags: -l 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

CPU Busy time: 5, IO Busy time:4. Stats: CPU Busy 5 (50.00%). Stats: IO Busy 4 (40.00%)

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16.0

#### Score 2:

16.0

#### Score 3:

16

## Entry: 3

### Item: 4

#### Question:

Now run with these flags: -I 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

It takes 10 time unit to complete both the processes.

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8.0

#### Score 2:

8.0

#### Score 3:

8

## Entry: 4

### Item: 4

#### Question:

Now run with these

flags: -I 4:100,1:0. These

flags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be completed. Can you find out if you were right.

#### Answer:

It takes Time 10 to complete both processes.

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE state.)

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16.0

#### Score 2:

8

#### Score 3:

8

## Entry: 5

### Item: 4

#### Question:

Now run with these

flags: -l 4:100,1:0. These

flags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be completed. Can you find out if you were right.

#### Answer:

It takes 9 time units to complete

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE state.)

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8

#### Score 2:

8

#### Score 3:

8

#### Score outlier:

0

## Entry: 6

### Item: 4

#### Question:

Now run with these flags: -I 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

It takes 10 time unit.

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8.0

#### Score 2:

8.0

#### Score 3:

8

## Entry: 7

### Item: 4

#### Question:

Now run with these flags: -I 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

It will take 10 ticks. Time PID: 0 PID: 1 CPU IOs 1 RUN:cpu READY 1 2 RUN:cpu READY 1 3 RUN:cpu READY 1 4 RUN

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16.0

#### Score 2:

16.0

#### Score 3:

16

#### Score outlier:

8

## Entry: 8

### Item: 4

#### Question:

Now run with these

flags: -l 4:100,1:0. These

flags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be completed. Can you find out if you were right.

#### Answer:

It takes 10 time units to complete both processes. Time PID: 0 PID: 1 CPU IOs 1 RUN:cpu READY 1 2 RUN:cpu READY

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE state.)

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16.0

#### Score 2:

16.0

#### Score 3:

16.0



## Entry: 9

### Item: 4

#### Question:

Now run with these

flags: -l 4:100,1:0. These

flags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be completed. Can you find out if you were right.

#### Answer:

9 time units

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE state.)

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8.0

#### Score 2:

8

#### Score 3:

8

## Entry: 10

### Item: 4

#### Question:

Now run with these

flags: -l 4:100,1:0. These

flags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be completed. Can you find out if you were right.

#### Answer:

9 time units. First processor 4:100 use 4 time units for running regular CPU instruction while second processor 1:0 issue

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16

#### Score 2:

16

#### Score 3:

16

## Entry: 11

### Item: 4

#### Question:

Now run with these \uFB01ags: -l 4:100,1:0. These \uFB01ags specify one process with 4 instructions (all to use the CPU

#### Answer:

It takes 10 ticks

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8.0

#### Score 2:

8.0

#### Score 3:

8

#### Score outlier:

16

## Entry: 12

### Item: 4

#### Question:

Now run with these

flags: -l 4:100,1:0. These

flags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be c

#### Answer:

It needs 10 unit time. hycheun456@ubt20a:~\$ python2 process-run.py -l 4:100,1:0 -c -p Time PID: 0 PID: 1 CPU IOs 1 R

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16.0

#### Score 2:

16.0

#### Score 3:

16

## Entry: 13

### Item: 4

#### Question:

Now run with these flags: -I 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

It takes for 10 times to complete both proesss

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8.0

#### Score 2:

8.0

#### Score 3:

8

## Entry: 14

### Item: 4

#### Question:

Now run with these flags: -l 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one

#### Answer:

It took 10 time units to finish both processes

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8

#### Score 2:

8

#### Score 3:

8

## Entry: 15

### Item: 4

#### Question:

Now run with these flags: -I 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

It took 10 time units to finish both processes

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8.0

#### Score 2:

8.0

#### Score 3:

8

## Entry: 16

### Item: 4

#### Question:

Now run with these flags: -l 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one

#### Answer:

The total time is 10. 4 CPU for the first process, 1 CPU and 4 I/O are waiting for the second process.

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16

#### Score 2:

16

#### Score 3:

16

#### Score outlier:

10



## Entry: 17

### Item: 4

#### Question:

Now run with these flags: -I 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

It takes 10 unit time to complete both processes.

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8.0

#### Score 2:

8

#### Score 3:

8

## Entry: 18

### Item: 4

#### Question:

Now run with these

flags: -l 4:100,1:0. These

flags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be completed. Can you find out if you were right.

#### Answer:

It takes 10 time units to complete. The first process that runs is in the RUNNING STATE which is using the CPU, while the second process is in the WAITING STATE.

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE state.)

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16

#### Score 2:

16

#### Score 3:

16

## Entry: 19

### Item: 4

#### Question:

Now run with these flags: -I 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

4 units for process 0 instructions, and then 6 units for process 1 issue an io and waiting. So, total 10 time units

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16.0

#### Score 2:

16.0

#### Score 3:

16

## Entry: 20

### Item: 4

#### Question:

Now run with these flags: -I 4:100,1:0. These flags specify one process with 4 instructions (all using the CPU), and one th

#### Answer:

The processes will take 10 clock cycles to complete the instructions as there are 4 instructions which take 1 cycle to comp

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16.0

#### Score 2:

16.0

#### Score 3:

16

# Entry: 21

## Item: 4

### Question:

Now run with these

■ags: -l 4:100,1:0. These

■ags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be

### Answer:

```
python2 ./process-run.py -l 4:100,1:0 -c
```

Time PID: 0 PID: 1 CPU IOs

1 RUN:cpu READY

1

2 RUN:cpu READY

1

3 RUN:cpu READY

1

4 RUN:cpu READY

1

5 DONE RUN:io-start

1

6 DONE WAITING

1

7 DONE WAITING

1

8 DONE WAITING

1

9 DONE WAITING

1

10 DONE DONE

Total time: 10

### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

### Sample criteria:

2 sub-questions: 8 points/each sub-question

### Full points:

16

### Score 1:

16

### Score 2:

16

### Score 3:



## Entry: 22

### Item: 4

#### Question:

Now run with these

flags: -l 4:100,1:0. These

flags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be done. Find out if you were right.

#### Answer:

It takes 9-time cycles to be done.

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE state.)

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8

#### Score 2:

8

#### Score 3:

8

## Entry: 23

### Item: 4

#### Question:

Answers

#### Answer:

8 time units

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE state.)

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

2

#### Score 2:

0

#### Score 3:

0



## Entry: 24

### Item: 4

#### Question:

Now run with these

flags: -l 4:100,1:0. These

flags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be c

#### Answer:

it takes 10 time unit to finish both processes. First 4 instructions each use 1 time unit, at the time unit 5, I/O start and wait

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16.0

#### Score 2:

16.0

#### Score 3:

16

## Entry: 25

### Item: 4

#### Question:

Now run with these fags: -l 4:100,1:0. These fags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

It takes 10 ticks to complete both processes since process 0 first use 4 ticks to run the CPU, process 1 uses 5 ticks to iss

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16.0

#### Score 2:

16.0

#### Score 3:

16

## Entry: 26

### Item: 4

#### Question:

Now run with these ■l 4:100,1:0. These ■lags specify one process with 4 instructions (all to use the CPU), and one that s

#### Answer:

There are 4 occurrences of &quot;cpu&quot; for Process 0 . The Process 1 starts with &quot;io-start&quot; and then trans

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

0.0

#### Score 2:

0.0

#### Score 3:

0

## Entry: 27

### Item: 4

#### Question:

Now run with these flags: -l 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one

#### Answer:

Time 1 to Time 4: The CPU process with PID 0 runs and uses the CPU for 4 time units. Time 5: The CPU process with PID

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16.0

#### Score 2:

16.0

#### Score 3:

16

## Entry: 28

### Item: 4

#### Question:

Now run with these flags: -l 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one

#### Answer:

$4 \times 100 = 400$  units of time

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

2

#### Score 2:

0

#### Score 3:

0

## Entry: 29

### Item: 4

#### Question:

Now run with these flags: -l 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one find out if you were right.

#### Answer:

The time of CPU is 5 and IO is 4. The total time is 10. Time PID: 0 PID: 1 CPU IOs 1 RUN:cpu READY 1 2 RUN:cpu REA

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16

#### Score 2:

16

#### Score 3:

16

## Entry: 30

### Item: 4

#### Question:

Now run with these flags: -I 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

For process 4:100, it takes 4 time units. For process 1:0, it takes 7 time units. In total, it takes 11 time units.

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

0.0

#### Score 2:

0.0

#### Score 3:

0

## Entry: 31

### Item: 4

#### Question:

Now run with these

flags: -l 4:100,1:0. These

flags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be completed. Can you find out if you were right.

#### Answer:

9 time units

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE state.)

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8.0

#### Score 2:

8

#### Score 3:

8



## Entry: 32

### Item: 4

#### Question:

Now run with these ■iags: -l 4:100,1:0. These ■iags specify one process with 4 instructions (all to use the CPU), and one

#### Answer:

It takes 10 units of time to complete these instructions.

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8

#### Score 2:

8

#### Score 3:

8

## Entry: 33

### Item: 4

#### Question:

Now run with these

flags: -I 4:100,1:0. These

flags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be c

#### Answer:

4 instructions of the first process are executed in 4 ticks, because they are CPU instructions. The one instruction of the se

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16.0

#### Score 2:

16.0

#### Score 3:

16

## Entry: 34

### Item: 4

#### Question:

Now run with these flags: -l 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

9-time units. Process 0 (4:100) takes 4-time units to execute its CPU instructions. Process 1 (1:0) takes 5-time units to co

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16.0

#### Score 2:

16.0

#### Score 3:

16

## Entry: 35

### Item: 4

#### Question:

Now run with these

flags: -l 4:100,1:0. These

flags specify one process with 4 instructions (all to use the CPU), and one that simply issues an I/O and waits for it to be completed. Can you find out if you were right.

#### Answer:

The total time is 10.

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE state.)

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16.0

#### Score 2:

8

#### Score 3:

8

## Entry: 36

### Item: 4

#### Question:

Now run with these flags: -I 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

10times

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8

#### Score 2:

8

#### Score 3:

8

## Entry: 37

### Item: 4

#### Question:

Now run with these flags: -I 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

The time is 10

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8

#### Score 2:

8

#### Score 3:

8

## Entry: 38

### Item: 4

#### Question:

Now run with these flags: -l 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

11 time units

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

10.0

#### Score 2:

0

#### Score 3:

0

## Entry: 39

### Item: 4

#### Question:

Now run with these flags: -l 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

10 clock ticks,

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

16

#### Score 2:

16

#### Score 3:

16



## Entry: 40

### Item: 4

#### Question:

Now run with these flags: -I 4:100,1:0. These flags specify one process with 4 instructions (all to use the CPU), and one th

#### Answer:

It takes 10 time-units to complete both processes.

#### Sample answer:

9 (or 10) time units.

Process 0 (4:100) uses four time units for running regular CPU instructions.

Process 1 (1:0) issues one I/O request, using five time units to complete. (It needs one more tick to switch to the DONE s

#### Sample criteria:

2 sub-questions: 8 points/each sub-question

#### Full points:

16

#### Score 1:

8.0

#### Score 2:

8.0

#### Score 3:

8